

AI-Powered Telehealth & Remote Patient Monitoring Platform

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PROJECT PROPOSAL: AI-POWERED TELEHEALTH & REMOTE PATIENT MONITORING PLATFORM

1. PROJECT OVERVIEW

We are seeking an experienced software development partner to build a HIPAA-compliant, AI-powered Telehealth and Remote Patient Monitoring (RPM) platform. The system will enable healthcare providers to conduct virtual consultations, continuously monitor patients' vital signs through connected medical devices (via Bluetooth/IoT), and leverage artificial intelligence to automate triage and patient interaction.

2. TARGET USERS

- Patients: Individuals needing virtual care and remote health monitoring.
- Clinicians/Doctors: Healthcare professionals conducting consultations, reviewing vitals, and prescribing care.
- Clinic Administrators: Staff managing scheduling, billing, user access, and reporting.

3. CORE FEATURES & FUNCTIONAL REQUIREMENTS

- HIPAA-Compliant Patient Portal (Web & Mobile): Secure login, profile management, medical history tracking, and consent form signing.
- Clinician Portal (Web-based Dashboard): Patient list, electronic health record (EHR) integration, vitals tracking charts, and prescribing tool.
- Real-time Video & Audio Consultations: High-quality, low-latency video calling directly in-app using WebRTC (or integrated via Twilio Programmable Video / Agora). Includes screen-sharing and in-call chat.
- AI-Powered Symptom Checker & Assistant: An integrated LLM-based chatbot that acts as a first-line support for patients to log symptoms, perform basic triage, and answer general health-related questions.
- Remote Patient Monitoring (IoT Integration): Integration with smart vitals monitoring devices (blood pressure cuff, pulse oximeter, glucometer) via BLE (Bluetooth Low Energy) for mobile, and API/Webhooks for cellular devices.
- Smart Scheduling & Reminders: Automated appointment scheduling, clinician availability calendar, SMS/Email/WhatsApp notification alerts for upcoming sessions and medication schedules.
- Secure Billing & Payment Gateway: Integration with Stripe for co-pays, subscription-based care management, and insurance details capture.
- EHR/EMR Integration: Integration with HL7/FHIR-compliant health record systems (such as Epic or Cerner) to sync patient files.
- Security & Compliance: End-to-end data encryption (AES-256 in transit and at rest), Multi-Factor Authentication (MFA), role-based access control, and complete audit logging of all system actions.

4. TECHNICAL SPECIFICATIONS & STACK PREFERENCES

- Frontend (Web): Next.js with React, TypeScript, Tailwind CSS, and Recharts for interactive vital trend visualization.
- Mobile App: Cross-platform mobile development using Flutter or React Native to support iOS and Android.
- Backend: Node.js with Express.js, TypeScript, and NestJS (preferred for architectural structure).

- Database: PostgreSQL (for relational patient records and audit logs) and Redis (for session caching and real-time chat message queues).
- Cloud Infrastructure: Amazon Web Services (AWS) utilizing ECS (Fargate) for container orchestration, RDS for managed PostgreSQL, S3 for secure document storage, Route53, and CloudFront.
- AI Integrations: OpenAI API (GPT-4o) or Anthropic Claude API for symptom checker reasoning.

5. ESTIMATION, BUDGET, AND TIMELINE

- Estimated Timeline: We expect the project to take approximately 5 to 6 months to complete a production-ready MVP.
- Budget Range: \$80,000 – \$110,000 USD.
- Key Milestones:
 - Phase 1: Discovery & UI/UX Design (Weeks 1-4) - User flows, wireframes, and high-fidelity interactive mockups.
 - Phase 2: Core Platform & HIPAA Infrastructure (Weeks 5-12) - Database schema, user auth, patient/doctor portal backbones, EHR/FHIR endpoints.
 - Phase 3: Video Calling & IoT/BLE Integration (Weeks 13-18) - WebRTC consultation implementation and bluetooth SDK sync for vital devices.
 - Phase 4: AI Assistant & Triage System (Weeks 19-22) - LLM integration, custom knowledge base routing, and auto-triage dashboard.
 - Phase 5: Quality Assurance & Launch (Weeks 23-26) - Security audits, HIPAA compliance review, load testing, and deployment to AWS production environments.